Advanced Research Methodology:
RFM-Based Customer Segmentation in the Banking Sector Using K-Means Clustering

Pattanasavich Meenandhavech
Agathiyan Akilan
Master's in Data Science
University of Europe for Applied Sciences, Potsdam Campus

May 20, 2026

Abstract

Customer segmentation is a critical analytical process in modern banking, enabling targeted marketing, risk assessment, and resource allocation. This paper outlines an advanced methodological framework for segmenting banking customers based on transactional behavior using Recency, Frequency, and Monetary (RFM) modeling combined with K-Means clustering. Utilizing a large-scale real-world dataset of banking transactions, we engineered RFM features relative to a deterministic historical baseline. To address the inherent power-law distributions in financial data, log-transformations and variance standardization were applied. Through mathematical evaluation using Within-Cluster Sum of Squares (Inertia) and the Silhouette Coefficient, an optimal configuration of $K = 5$ was selected, yielding distinct and highly actionable banking personas.

---

1 Introduction

In the highly competitive financial sector, understanding customer behavior is paramount for maximizing customer lifetime value (CLV) and minimizing churn. Traditional demographic segmentation often fails to capture the nuanced interaction frequencies and financial volumes of modern banking clients. Behavioral segmentation, particularly the Recency, Frequency, and Monetary (RFM) model, provides a deterministic lens into customer engagement.

This research project applies K-Means clustering—an unsupervised machine learning algorithm—to an engineered RFM matrix. The objective is to identify distinct, homogenous groups of customers within a real-world banking dataset to drive strategic business initiatives.

2 Data and Preprocessing Methodology

2.1 Dataset Description

The analysis utilizes the "Bank Customer Segmentation" dataset, featuring raw transactional logs. Key attributes extracted include `CustomerID`, `TransactionDate`, and `TransactionAmount (INR)`. Relying on raw logs rather than pre-aggregated data allows for strict methodological control over feature engineering and temporal baselining.

2.2 Feature Engineering (RFM)

The structural RFM metrics were mathematically derived for each unique `CustomerID`:

- **Recency (R):** Days elapsed since the customer's last transaction relative to a standardized snapshot date (T_{snapshot}), defined as one day past the most recent historical log in the dataset.

$$R_i = T_{\text{snapshot}} - \max(T_{i,\text{tx}}) \tag{1}$$

- **Frequency (F):** Total count of discrete transactions executed by the customer.

$$F_i = \sum(\text{Transactions}_i) \tag{2}$$

- **Monetary (M):** The aggregated financial volume (in INR) moved by the customer.

$$M_i = \sum(\text{TransactionAmount}_{i,\text{tx}}) \tag{3}$$

2.3 Statistical Transformation and Normalization

Banking data inherently exhibits extreme variance imbalance and right-skewness (power-law distribution). Because K-Means utilizes Euclidean distance, these raw structural anomalies would mathematically distort the cluster boundaries, causing the *Monetary* dimension to monopolize the distance calculations.

To mitigate this, a logarithmic transformation, $\log(x+1)$, was applied to compress the heavy right-skewness and safely handle zero-bounds. Subsequently, a Standard Scaler was applied to center the feature matrix to a mean of zero and a variance of one ($\mu = 0, \sigma^2 = 1$). This ensures isotropic cluster formation across all three dimensions.

3 Model Evaluation and Selection of K

Determining the optimal number of clusters (K) requires balancing mathematical rigor with business utility. The algorithm was evaluated across $K \in [2, 6]$ using two primary metrics:

1. **Inertia (WCSS):** Measures cluster cohesion.
2. **Silhouette Coefficient:** Measures cluster separation and boundary overlap.

Table 1: Structural Optimization Metrics Evaluation

Clusters (K)	**Inertia (WCSS)**	**Silhouette Score**
2	1.716×10^6	0.5187
3	1.271×10^6	0.3498
4	9.643×10^5	0.3626
5	7.450×10^5	**0.3731**
6	6.211×10^5	0.3669

While $K = 2$ achieved the global mathematical maximum for the Silhouette Score (0.5187), this binary split (Active vs. Inactive) lacks strategic depth for financial applications. Conversely, $K = 5$ represents an optimal localized peak in the Silhouette distribution (0.3731) coinciding with the stabilization of the Inertia curve (the "Elbow"). Therefore, $K = 5$ was selected for the final operational model.

4 Final Segment Profiling and Business Personas

The K-Means model ($K = 5$) was mapped back to the unscaled, real-world monetary and temporal metrics, yielding five strategically distinct banking personas.

Table 2: K-Means Clustering: Final Banking Personas

Cluster	**Persona Assignment**	**Behavioral Profile & Strategic Implication**
0	**The Wealth Champions**	*Low Recency, Med-High Frequency, High Monetary.* High-net-worth active accounts moving massive capital volumes frequently. Target with premium wealth management perks.
1	**Dormant / At-Risk Users**	*High Recency, Low Frequency, Low-Med Monetary.* Churned or inactive customers with long gaps since last transaction. Target with re-engagement campaigns.
2	**High-Velocity Retail**	*Low Recency, High Frequency, Low Monetary.* Daily retail transactors for low-value payments. High engagement, low margin. Target with cross-sell credit products.
3	**Occasional Investors**	*Med Recency, Low Frequency, High Monetary.* Users transferring large lump sums infrequently (e.g., term depositors, salary-dump). Target with fixed deposit options.
4	**Lost / Low-Value**	*Med-High Recency, Low Frequency, Low Monetary.* Low engagement and low volume, presenting maintenance overhead. Deploy low-cost automated retention triggers.

5 Conclusion

This framework successfully demonstrates that behavioral transactional data, when properly standardized and mathematically evaluated via K-Means clustering, yields highly robust and strategically relevant customer segments. The extraction of five distinct personas from raw unclassified logs validates the utility of RFM optimization in modern data science methodologies, providing actionable pathways for targeted resource allocation in the banking sector.